MILITARY SPECIFICATION

CABLE ASSEMBLIES, FOR M91A1 ADAPTION KIT

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for a series of sixteen electrical cable assemblies and three jumper plugs designated as Cable Assemblies. The identification of the Cables are W1L, W2L, W3L, W5L, W6L, W7L, W8L, W9L, W10L, W25L, W44L, T4138-1, T4138-3, T4138-4 and T4138-5. The identifications of the jumper plugs are Check Plug - 2P6, Fire Plug - 8830734, and Safe Plug - 8830733, which are used in the Sergeant warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for.

FSC 1135

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DRAWINGS

ORDANCE CORPS

8830073 - Cable Assembly, T.D.D. (W2L).
8830116 - Cable, Impact Fuze, Forward (W7L).
8830118 - Cable Assembly, T.D.D. (W1L).
8830119 - Cable, Fuze (W6L).
8830126 - Cable Assembly, Baro (W3L).
8830171 - Harness, Parameter Insertion (W5L).
8830298 - Cable, Fuze (W9L).
8831299 - Cable, Fuze (W8L).
8830272 - Cable Assembly, Test Set (T4138-5).
8830273 - Cable Assembly, Special Purpose, Electrical Branched (CU) (T4138-4).
8830465 - Cable Assembly, Special Purpose, Electrical (P919) (T4138-3).
8830473 - Cable Assembly, Power, Electrical (AC) (T4138-1).
8830306 - Check Plug.
8830733 - Safe Plug.
8830734 - Fire Plug.
8847860 - Cable Assembly, Power Electrical (P.S.) (T4138-2).
8862811 - Cable, Extension (W25L).
8863034 - Cable, Warhead (W44L).
8865045 - Cable Assembly, Special Purpose Electrical (E10L).
EL-8854164 - Equipment List.

PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The cable assemblies shall comply with all requirements specified on the applicable drawing, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

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3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

3.6 Mating assembly. The cable assemblies shall be capable of being completely mated and unmated with their proper receptacles, without use of tools. This capability shall be verified as specified in 4.4.1.

3.7 Circuit resistance. The resistance between the pins of the cable assemblies listed in Tables I through XVIII shall be as specified when tested in accordance with 4.4.3.

Table I

T.D.D. cable assemblies (W1L) and (W2L) continuity test

From	To	Resistance (Ohms)
P1-E	P2-C	0.10 max.
P1-F	P2-D	0.10 max.
P1-G	P2-E	0.10 max.
P1-H	P2-F	0.10 max.
P1-J	P2-G	0.10 max.
P1-K	P2-H	0.10 max.
P1-L	P2-J	0.10 max.
P1-M	P2-K	0.10 max.
P1-N	P2-L	0.10 max.
P1-V	P2-N	0.10 max.
P1-a	P2-T	0.10 max.
P1-b	P2-U	0.10 max.
P1-c	P2-V	0.10 max.
P1-d	P2-W	0.10 max.
P1-e	P2-X	0.10 max.
P1-f	P2-R	0.10 max.
P1-g	P2-S	0.10 max.
P1-h	P2-Y	0.10 max.
P1-i	P2-Z	0.10 max.
P1-j	P2-a	0.10 max.
P1-k	P2-b	0.10 max.
P1-m	P2-c	0.10 max.
P1-n	P2-d	0.10 max.

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P1-r	P2-e	0.10 max.
P1-s	P2-f	0.10 max.
P1-t	P2-g	0.10 max.
P1-D	P2-B	0.10 max.

Table II

Baro cable (W3L) continuity test

From	To	Resistance (Ohms)
P1-A	P2-A	0.1 max.
P1-B	P2-B	0.1 max.
P1-C	P2-C	0.1 max.
P1-D	P2-D	0.1 max.
P1-E	P2-E	0.1 max.
P1-F	P2-F	0.1 max.
P1-G	P2-G	0.1 max.
P1-H	P2-H	0.1 max.
P1-J	P2-J	0.1 max.
P1-K	P2-K	0.1 max.
P1-L	P2-L	0.1 max.
P1-M	P2-M	0.1 max.
P1-N	P2-N	0.1 max.
P1-P	P2-P	0.1 max.
P1-R	P2-R	0.1 max.
P1-S	P2-S	0.1 max.
P1-T	P2-T	0.1 max.
P1-U	P2-U	0.1 max.
P1-V	P2-V	0.1 max.
P1-W	P2-W	0.1 max.
P1-Z	P2-Z	0.1 max.
P1-b	P1-b	0.1 max.
P1-c	P1-c	0.1 max.

Table III

Parameter insertion harness (W5L) continuity test

From	To	Resistance (Ohms)
P1-A	P2-c	0.2 max.
P1-B	P2-X	0.2 max.
P1-C	P919-14	0.2 max.
P1-D	P919-21	0.2 max.
P1-E	P919-3	0.2 max.
P1-F	P919-1	0.2 max.
P1-G	P919-69	0.2 max.
P1-H	P919-50	0.2 max.

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P1-J	P919-8	0.2 max.
P1-K	P2-d	0.2 max.
P1-L	P2-$\overline{Z}$	0.2 max.
P1-M	P919-70	0.2 max.
P1-N	P2-W	0.2 max.
P1-R	P919-27	0.2 max.
P1-S	P2-b	0.2 max.
P1-T	P919-35	0.2 max.
P1-U	P919-41	0.2 max.
P1-V	P2-Y	0.2 max.
P1-X	P919-22	0.2 max.
P1-Y	P919-11	0.2 max.
P1-Z	P2-L	0.2 max.
P1-a	P919-40	0.2 max.
P1-$\overline{b}$	P919-15	0.2 max.
P1-$\overline{c}$	P919-13	0.2 max.
P1-$\overline{e}$	P919-53	0.2 max.
P1-$\overline{f}$	P919-36	0.2 max.
P1-$\overline{r}$	J1-B	0.2 max.
P1-$\overline{p}$	J1-A	0.2 max.
P1-$\overline{n}$	J1-C	0.2 max.
P1-$\overline{E}$	P2-C	0.2 max.
P1-Y	P2-D	0.2 max.
P1-d	P2-K	0.2 max.
P1-$\overline{W}$	P2-E	0.2 max.
P1-m	P919-18	0.2 max.
P1-$\overline{m}$	P2-P	0.2 max.
P1-$\overline{h}$	P919-42	0.2 max.
P1-$\overline{j}$	P919-5	0.2 max.
P1-$\overline{i}$	P919-63	0.2 max.
P2-$\overline{S}$	P919-9	0.2 max.
P2-F	P919-31	0.2 max.
P2-H	P919-68	0.2 max.
P2-N	P919-17	0.2 max.
P2-R	P919-62	0.2 max.
P2-J	P919-28	0.2 max.
P2-G	P919-24	0.2 max.
P2-M	P919-4	0.2 max.

Table IV

Fuze cable (W6L) continuity test

From	To	Resistance (Ohms)
P1-A	P3-P	0.10 max.
P1-B	P3-Z	0.10 max.
P1-C	P3-G	0.10 max.
P1-D	P3-J	0.10 max.

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P1-E	P3-L	0.10 max.
P1-F	P3-C	0.10 max.
P1-G	P3-V	0.10 max.
P1-H	P3-X	0.10 max.
P1-J	P3-Y	0.10 max.
P1-K	P2-J	0.10 max.
P1-L	P2-F	0.10 max.
P1-M	P2-K	0.10 max.
P1-N	P2-C	0.10 max.
P1-P	P2-D	0.10 max.
P1-R	P4-B	0.10 max.
P1-S	P4-C	0.10 max.
P1-T	P4-F	0.10 max.
P1-U	P4-H	0.10 max.
P1-V	P2-B	0.10 max.
P1-W	P4-L	0.10 max.
P1-Z	P2-A	0.10 max.
P1-b	P2-E	0.10 max.
P2-$\overline{P}$	P4-A	0.10 max.
P3-A	P3-W	0.10 max.
P3-B	P3-S	0.10 max.
P3-E	P3-H	0.10 max.
P3-F	P3-M	0.10 max.
P3-N	P3-R	0.10 max.
P3-T	P3-U	0.10 max.
P4-D	P4-E	0.10 max.
P4-G	P4-M	0.10 max.
P4-J	P4-K	0.10 max.

Table V

Forward impact fuze cable (W7L) continuity test

From	To	Resistance (Ohms)
P1-Center	P2-Center	0.1 max.
P1-Center	P3-Center	0.1 max.
P1-Shell	P2-Shell	0.1 max.
P1-Shell	P3-Shell	0.1 max.

Table VI

Fuze cable (W8L) continuity test

From	To	Resistance (Ohms)
P1-Center	P2-Center	0.1 max.
P1-Center	P3-Center	0.1 max.
P1-Center	P4-Center	0.1 max.

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P1-Shell	P2-Shell	0.1 max.
P1-Shell	P3-Shell	0.1 max.
P1-Shell	P4-Shell	0.1 max.

Table VII

Fuze cable (W9L) continuity test

From	To	Resistance (Ohms)
P1-Center	P2-Center	0.1 max.
P1-Center	P3-Center	0.1 max.
P1-Center	P4-Center	0.1 max.
P1-Center	P5-Center	0.1 max.
P1-Shell	P2-Shell	0.1 max.
P1-Shell	P3-Shell	0.1 max.
P1-Shell	P4-Shell	0.1 max.
P1-Shell	P5-Shell	0.1 max.

Table VIII

Cable assembly (W10L) continuity test

From	To	Resistance (Ohms)
P1-Shell	P2-Shell	0.10 max.
P1-A	P2-A	0.10 max.
P1-B	P2-B	0.10 max.
P1-C	P2-C	0.10 max.
P1-D	P2-D	0.10 max.
P1-E	P2-E	0.10 max.
P1-F	P2-F	0.10 max.
P1-G	P2-G	0.10 max.
P1-H	P2-H	0.10 max.
P1-J	P2-J	0.10 max.
P1-K	P2-K	0.10 max.
P1-L	P2-L	0.10 max.
P1-M	P2-M	0.10 max.
P1-U	P2-U	0.10 max.
P1-P	P2-P	0.10 max.
P1-R	P2-R	0.10 max.
P1-S	P2-S	0.10 max.

Table IX

Warhead cable (W44L) continuity test

From	To	Resistance (Ohms)
P1-U	P2-D	0.15 max.

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P1-V	P2-E	0.15 max.
P1-e	P2-A	0.15 max.
P1-r	P2-N	0.15 max.
P1-H	P2-J	0.15 max.
P1-n	P2-W	0.15 max.
P1-k	P2-X	0.15 max.
P1-L	P2-F	0.15 max.
P1-M	P2-G	0.15 max.
P1-m	P3-B	0.15 max.
P1-s	P3-A	0.10 max.
P1-C	P3-D	0.15 max.
P1-p	P4-B	0.15 max.
P1-q	P4-A	0.10 max.
P1-J	P4-E	0.15 max.
P1-E	P2-T	0.15 max.
P1-G	P2-P	0.15 max.
P1-D	P2-R	0.15 max.
P1-K	P2-K	0.15 max.
P1-R	P2-S	0.10 max.
P1-P	P2-V	0.10 max.
P3-E	P4-D	0.10 max.
P1-S	P2-H	0.10 max.

Table X

Extension cable (W25L) continuity test

From	To	Resistance (Ohms)
P1-B	P2-B	0.1 max.
P1-E	P2-E	0.1 max.
P1-G	P2-G	0.1 max.
P1-K	P2-K	0.1 max.
P1-r	P2-r	0.1 max.
P1-e	P2-e	0.1 max.
P1-V	P2-V	0.1 max.
P1-U	P2-U	0.1 max.
P1-H	P2-H	0.1 max.
P1-n	P2-n	0.1 max.
P1-k	P2-k	0.1 max.
P1-M	P2-M	0.1 max.
P1-L	P2-L	0.1 max.
P1-D	P2-D	0.1 max.
P1-S	P2-S	0.1 max.
P1-R	P2-R	0.1 max.
P1-P	P2-P	0.1 max.
P1-m	P2-m	0.1 max.
P1-s	P2-s	0.1 max.
P1-C	P2-C	0.1 max.

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P1-p	P2-p	0.1 max.
P1-q	P2-q	0.1 max.
P1-j	P2-j	0.1 max.

Table XI

Cable, power, electrical (AC) (T-4138-1) continuity test

From	To	Resistance (Ohms)
P1-A	P2-1	0.10 max.
P1-C	P2-2	0.10 max.
P1-B	P2-Ground	0.10 max.

Table XII

Cable assembly, power, electrical (PS) (T-4138-2) continuity test

From	To	Resistance (Ohms)
P1-A	P2-A	0.2 max.
P1-C	P2-C	0.2 max.
P1-F	P2-F	0.2 max.
P1-G	P2-G	0.2 max.
P1-H	P2-H	0.2 max.
P1-K	P2-K	0.2 max.
P1-L	P2-L	0.2 max.
P1-M	P2-M	0.2 max.
P1-T	P2-T	0.2 max.
P1-U	P2-U	0.2 max.
P1-V	P2-V	0.2 max.
P1-W	P2-W	0.2 max.
P1-Y	P2-Y	0.2 max.
P1-Z	P2-Z	0.2 max.
P1-a	P2-a	0.2 max.
P1-e	P2-e	0.2 max.
P1-f	P2-f	0.2 max.
P1-g	P2-g	0.2 max.
P1-h	P2-h	0.2 max.
P1-i	P2-i	0.2 max.
P1-j	P2-j	0.2 max.
P1-k	P2-k	0.2 max.
P1-p	P2-p	0.2 max.
P1-q	P2-q	0.2 max.
P1-r	P2-r	0.2 max.
P1-s	P2-s	0.2 max.
P1-u	P2-u	0.2 max.

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P1-v	P2-v	0.2 max.
P1-w	P2-w	0.2 max.
P1-x	P2-x	0.2 max.
P1-y	P2-y	0.2 max.
P1-z	P2-z	0.2 max.
P1-AA	P2-AA	0.2 max.
P1-BB	P2-BB	0.2 max.
P1-DD	P2-DD	0.2 max.
P1-EE	P2-EE	0.2 max.
P1-FF	P2-FF	0.2 max.
P1-GG	P2-GG	0.2 max.
P1-HH	P2-HH	0.2 max.
P1-JJ	P2-JJ	0.2 max.
P1-KK	P2-KK	0.2 max.
P1-LL	P2-LL	0.2 max.
P1-MM	P2-MM	0.2 max.
P1-NN	P2-NN	0.2 max.
P1-PP	P2-PP	0.2 max.

Table XIII

Cable assembly, special purpose, electrical: (P919) (T-4138-3)

From	To	Resistance (Ohms)
P1-B	J919-24	0.2 max.
P1-C	J919-3	0.2 max.
P1-D	J919-18	0.2 max.
P1-E	J919-69	0.2 max.
P1-F	J919-50	0.2 max.
P1-G	J919-62	0.2 max.
P1-H	J919-68	0.2 max.
P1-J	J919-13	0.2 max.
P1-K	J919-70	0.2 max.
P1-L	J919-31	0.2 max.
P1-M	J919-17	0.2 max.
P1-N	J919-35	0.2 max.
P1-P	J919-11	0.2 max.
P1-R	J919-4	0.2 max.
P1-S	J919-36	0.2 max.
P1-U	J919-14	0.2 max.
P1-V	J919-21	0.2 max.
P1-W	J919-1	0.2 max.
P1-X	J919-8	0.2 max.
P1-Y	J919-27	0.2 max.
P1-Z	J919-41	0.2 max.

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P1-a	J919-28	0.2 max.
P1-b	J919-22	0.2 max.
P1-c	J919-40	0.2 max.
P1-d	J919-15	0.2 max.
P1-e	J919-53	0.2 max.
P1-f	J919-9	0.2 max.
P1-g	J919-42	0.2 max.
P1-h	J919-63	0.2 max.
P1-j	J919-5	0.2 max.

Table XIV

Cable assembly, special purpose, electrical, branched (CU) (T-4138-4) continuity test

From	To	Resistance (Ohms)
P1-C	P2-v	0.2 max.
P1-E	J1-d	0.2 max.
P1-F	J1-p	0.2 max.
P1-G	P2-G	0.2 max.
P1-H	J1-h	0.2 max.
P1-J	J1-W	0.2 max.
P1-K	P2-K	0.2 max.
P1-L	J1-BB	0.2 max.
P1-M	P2-M	0.2 max.
P1-N	J1-g	0.2 max.
P1-R	P2-R	0.2 max.
P1-I	J1-j	0.2 max.
P1-V	J1-r	0.2 max.
P1-W	P2-W	0.2 max.
P1-Y	J1-a	0.2 max.
P1-Z	P2-Z	0.2 max.
P1-a	P2-a	0.2 max.
P1-b	P2-b	0.2 max.
P1-c	P2-c	0.2 max.
P1-d	P2-d	0.2 max.
P1-h	P2-h	0.2 max.
P1-i	P2-i	0.2 max.
P1-j	P2-j	0.2 max.
P1-m	J1-t	0.2 max.
P1-n	P2-n	0.2 max.
P1-p	P2-p	0.2 max.
P1-r	P2-r	0.2 max.
P1-s	P2-s	0.2 max.
P1-t	P2-t	0.2 max.
P1-u	P2-u	0.2 max.
P1-w	J1-GG	0.2 max.
P1-x	P2-x	0.2 max.
P1-y	P2-y	0.2 max.
P1-z	P2-z	0.2 max.

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P1-AA	J1-EE	0.2 max.
P1-BB	P2-BB	0.2 max.
P1-CC	P2-CC	0.2 max.
P1-DD	P2-DD	0.2 max.
P1-EE	P2-EE	0.2 max.
P1-FF	P2-FF	0.2 max.
P1-GG	P2-GG	0.2 max.
P1-HH	P2-HH	0.2 max.
P1-D	P3-V	0.2 max.
P1-g	J1-S	0.2 max.
P1-P	P3-H	0.2 max.
P1-X	P3-n	0.2 max.
P1-e	P3-K	0.2 max.
P1-f	P3-L	0.2 max.
P1-k	P3-M	0.2 max.
P1-U	P3-P	0.2 max.
P1-S	P3-S	0.2 max.
P1-q	P3-B	0.2 max.
P1-v	P3-b	0.2 max.
P1-A	P3-K	0.2 max.
P1-B	P3-U	0.2 max.
J1-f	P3-G	0.2 max.
J1-m	P2-m	0.2 max.
J1-e	P3-j	0.2 max.
J1-P	P2-P	0.2 max.
J1-N	P2-N	0.2 max.
J1-AA	P2-AA	0.2 max.
J1-w	P2-w	0.2 max.
J1-Y	P2-Y	0.2 max.
J1-V	P2-V	0.2 max.
J1-T	P2-T	0.2 max.
J1-L	P2-L	0.2 max.
J1-J	P2-J	0.2 max.
J1-H	P2-H	0.2 max.
J1-F	P2-F	0.2 max.
J1-E	P2-E	0.2 max.
P2-S	P2-q	0.05 max.
P2-e	P2-f	0.05 max.
P3-e	P3-r	0.05 max.

Table XV

Cable assembly, special purpose, electrical branched (fuze) (T-4138-5) continuity test

From	To	Resistance (Ohms)
P1-A	P3-u	0.2 max.
P1-B	P3-t	0.2 max.

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P1-C	P3-a	0.2 max.
P1-D	P3-HH	0.2 max.
P1-E	P3-AA	0.2 max.
P1-F	P3-EE	0.2 max.
P1-a	P3-e	0.2 max.
P1-p	P3-Y	0.2 max.
P1-V	P3-N	0.2 max.
P1-W	P3-M	0.5 max.
P1-W	P3-L	0.5 max.
P1-V	P3-P	0.5 max.
P1-V	P3-R	0.5 max.
P1-V	P3-S	0.5 max.
P1-V	P3-T	0.5 max.
P1-V	P3-U	0.5 max.
P1-V	P3-W	0.5 max.
P1-V	P3-X	0.5 max.
P1-BB	P3-h	0.5 max.
P1-AA	P3-V	0.2 max.
P1-m	P3-p	0.2 max.
P1-S	P3-BB	0.2 max.
P1-K	P2-u	0.2 max.
P1-L	P2-t	0.2 max.
P1-Z	P3-y	0.2 max.
P1-x	P2-x	0.2 max.
P1-M	P2-a	0.2 max.
P1-GG	P3-EE	0.5 max.
P1-N	P2-HH	0.2 max.
P1-P	P2-AA	0.2 max.
P1-DD	P2-EE	0.5 max.
P1-j	P2-e	0.2 max.
P1-y	P2-v	0.2 max.
P1-v	P2-Y	0.2 max.
P1-S	P3-v	0.2 max.
P1-e	P2-N	0.2 max.
P1-f	P2-M	0.5 max.
P1-f	P2-L	0.5 max.
P1-e	P2-P	0.5 max.
P1-e	P2-R	0.5 max.
P1-e	P2-S	0.5 max.
P1-e	P2-T	0.5 max.
P1-e	P2-U	0.5 max.
P1-e	P2-W	0.5 max.
P1-e	P2-X	0.5 max.
P1-u	P2-V	0.2 max.
P1-c	P2-p	0.2 max.
P1-G	P2-BB	0.2 max.
P1-i	P2-y	0.2 max.
P3-Z	P2-Z	0.2 max.
P1-r	P3-x	0.2 max.

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P1-R	P2-EE	0.2 max.
P2-h	P3-h	0.2 max.
P1-b	P3-h	0.2 max.
P1-z	P3-h	0.2 max.
P1-HH	P3-h	0.2 max.
P1-t	P2-h	0.2 max.
P1-k	P2-h	0.2 max.

Table XVI

Check plug 2P6 continuity test

From Pin	To Pin	Resistance (Ohms)
E	d	0.1 max.
H	h	0.1 max.
Y	a	0.1 max.
L	BB	0.1 max.
F	p	0.1 max.
J	W	0.1 max.
w	GG	0.1 max.
AA	EE	0.1 max.
N	g	0.1 max.
t	m	0.1 max.
e	f	0.1 max.
T	l	0.2 max.
V	r	0.2 max.
P	S	0.2 max.

Table XVII

Fire plug - 8830734 continuity test

Refer to dwg. 8830734 for Continuity Requirements

Table XVIII

Safe plug - 8830733 continuity test

None required.

3.8 Insulation resistance. The insulation resistance between ground and each terminal and all other terminals listed for the respective assemblies in Table XIX through XXXIV shall be greater than 20 megohms when tested at 500 ± 50 vdc as specified in 4.4.2. Unused terminals shall be connected together and tested as one terminal except for Check Plug 2P6, where the unused terminals shall be individually tested for isolation. Cable connector shells must be at ground potential (test fixture ground).

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Table XIX

T.D.D. cable assemblies (W1L and W2L)
insulation resistance test

P1-D, E, F, G, H, J, K, L, M, V, a, b, c, d, e, f, g, h, i, j, k, m, n, r, s, t.
P1-Unused terminals: A, B, C, P, R, S, T, U, W, X, Y, Z, p, q.
P2-Unused terminals: A, P, M.

Table XX

Baro cable (W3L)
insulation resistance test

P1-A, B, C, D, E, F, G, H, J, K, L, M, N, P, R, S, T, U, V, W, Z, b, c.
P1-Unused terminals: X, Y, a.
P2-Unused terminals: X, Y, a.

Table XXI

Parameter insertion harness (W5L)
insulation resistance test

P1-A, B, C, D, E, F, G, H, J, K, L, M, N, R, S, T, U, V, W, X, Y, Z, a, b, c, d, e, f, h, i, j, m, p, r, n.
P919- 4, 9, 17, 24, 28, 31, 62, 28.
P1-Unused terminals: P, g, q, k.
P2-Unused terminals: A, B, T, U, V, a, e, f, g.

P919-Unused terminals: 2, 6, 7, 10, 12, 16, 19, 20, 23, 25, 26, 29, 30, 32, 33, 34, 37, 38, 39, 43, 44, 45, 46, 47, 48, 49, 51, 52, 54, 55, 56, 57, 58, 59, 60, 61, 64, 65, 66, 67, 71, 72, 73.

Table XXII

Fuze cable (W6L)
insulation resistance test

P1-A, B, C, D, E, F, G, H, J, K, L, M, N, P, R, S, T, U, V, W, Z, b, c.
P3-A, E, F, N, T, S.
P4-D, G, J, A.
P1-Unused terminals: X, Y, a.
P2-Unused terminals: G, H, L, M, N, R, S, T, U, V.
P3-Unused terminals: D,K.
P4-Unused terminals: N, P, R.

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Table XXIII

Forward impact fuze cable (W7L)
fuze cable (W8L)
fuze cable (W9L)
insulation resistance test

P1-Conductor

Table XXIV

Cable assembly (W10L)
insulation resistance test

P1-Shell, A, B, C, D, E, F, G, H, J, K, L, M, U, P, R, S.
Unused Terminals: P1-N, T, and P2-N, T.

Table XXV

Warhead cable (W44L)
insulation resistance test

P1-C, D, E, G, H, K, L, M, P, R, S, U, V, e, j, k, m, n, p, q, r, s.
P3-E.
P1-Unused terminals: A, B, F, J, N, T, W, X, Y, Z, a, b, c, d, f, g, h, i, t.
P2-Unused terminals: B, C, L, M, U.
P3-Unused terminals: C,F.
P4-Unused terminals: C, F.

Table XXVI

Cable (W25L)
insulation resistance test

P1-E, G, K, r, e, V, U, H, n, k, M, L, D, S, R, P, m, s, C, p, q, j, B.
Unused Terminals: P1-h, Y, t, F, A, Z, N, J, T, W, a, d, f, g, i, b, c.
Unused Terminals: P2-h, Y, t, F, A, Z, N, J, T, W, X, a, d, f, g, i, b, c.

Table XXVII

Cable power, electrical (AC) (T-4138-1)
insulation resistance test

P1-A, B, C.
Ground - This will be the test fixture ground. Cable connectors shall be at ground potential.

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Table XXVIII

Cable assembly, power, electrical (PS) (T-4138-2)
insulation resistance test

P1-A, C, F, G, H, K, L, M, T, U, V, W, Y, Z, a, e, f, g, h, i, j, k, p, q, r, s, u, v, w, x, y, z, AA, BB, DD, EE, FF, GG, HH, JJ, KK, LL, MM, NN, PP.
P1-Unused terminals: B, D, E, J, N, P, R, S, X, b, c, d, m, n, t, CC.
P2-Unused terminals: B, D, E, J, N, P, R, S, X, b, c, d, m, n, t, CC.
Do not check shields M, K, a, k, v, BB, HH, LL and PP against each other, but check these shields against all other conductors and ground.

Table XXIX

Cable assembly, special purpose, electrical (P919) (T-4138-3)
insulation resistance test

P1-B, C, D, E, F, G, H, J, K, L, M, N, P, R, S, U, V, W, X, Y, Z, a, b, c, d, e, f, g, h, j.
P1-Unused terminals: A, T.
J919-Unused terminals: 2, 6, 7, 10, 12, 16, 19, 20, 23, 25, 26, 29, 30, 32, 33, 34, 37, 38, 39, 43, 44, 45, 46, 47, 48, 49, 51, 52, 54, 55, 56, 57, 58, 59, 60, 61, 64, 65, 66, 67, 71, 72, 73.

Table XXX

Cable assembly, special purpose, electrical,
branched (CU) (T-4138-4)
insulation resistance test

P1-A, B, C, D, E, F, G, H, J, K, L, M, N, P, R, S, T, U, V, W, X, Y, Z, a, b, c, d, e, f, g, h, i, j, k, m, n, p, q, r, s, t, u, v, w, x, y, z, AA, BB, CC, DD, EE, FF, GG, HH.
J1-E, F, H, J, L, N, P, T, V, Y, e, f, m, w, AA.
P3-e.
P2-S, e.
J1-Unused terminals: A, B, C, D, G, K, M, R, U, X, Z, b, c, i, k, n, q, s, u, v, x, y, z, CC, DD, FF, HH.
P3-Unused terminals: A, C, D, E, F, J, N, R, T, W, X, Y, Z, a, c, d, f, g, h, i, m, p, q, s, t.
P2-Unused terminals: A, B, C, D, U, X, f, k, q.

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Table XXXI

Cable assembly, special purpose, electrical, branched, (fuze) (T-4138-5) insulation resistance test

P1-A, B, C, D, E, F, G, K, L, M, N, P, R, S, V, W, Z, a, c, e, f, i, j, m, p, r, s, u, v, x, y, AA, BB.
P3-Z.
P1-Unused terminals: H, J, T, U, X, Y, CC, d, g, h, n, q, w, EE, FF.
P2-Unused terminals: A, B, C, D, E, F, G, H, J, K, b, c, d, f, g, i, j, k, m, n, q, r, s, w, CC, DD, FF, GG, z.
P3-Unused terminals: A, B, C, D, E, F, G, H, J, K, b, c, d, f, g, i, j, k, m, n, q, r, s, w, z, CC, DD, FF, GG.

Table XXXII

Check plug - 2P6 insulation resistance test

Pins E, F, H, J, L, t, Y, w, AA, e, N, P, T, V.
P1-Unused terminals: A, B, C, D, G, K, M, R, U, X, Z, b, c, i, k, m, q, s, u, v, x, y, z, CC, DD, FF, HH.

Table XXXIII

Fire plug - 8830734 insulation resistance test

Refer to 8830734 for insulation resistance requirements.

Table XXXIV

Safe plug - 8830733 insulation resistance test

Pins-A, B, C, D, E, F, G, H, J, K, L, M, N, P, R, S, T, U, V, W, X, Y, Z.

3.8.1 Diode characteristics. The diode characteristics shall meet the requirements of Table XXXV when tested as specified in 4.4.3.1.

Table XXXV

Fire plug 8830734 diode characteristics requirements

Refer to 8830734 for diode characteristics requirements.

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Positive Lead to Terminal	Negative Lead to Terminal	Applied Forward Current	Max. Forward Voltage Drop	Test Code Nos.
J2-H	J2-C	400 ± 20 ma	1.20	051001

Positive Lead to Terminal	Negative Lead to Terminal	Applied Reverse Voltage	Max. Volt Drop Across Series Resistor	Test Code Nos.
J2-C	J2-H	225 ± 12	4.0	051002

3.9 Environmental testing. The cable assemblies shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The cable assemblies shall also comply with 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.4. Tests shall be conducted in the order specified.

3.9.1 High temperature. The cables and plugs shall show no evidence of damage or deterioration such as distortion, cracks, or swelling when subjected to + 160 degrees Fahrenheit (°F.) + 5°F. as specified in 4.4.4.1 and shall meet the applicable requirements of 3.8 while at this temperature. Cables and plugs shall meet the applicable requirements of 3.7 after completion of high temperature test while at 80° + 20°F.

3.9.2 Low temperature. The cable assemblies shall show no evidence of damage or deterioration such as distortion, cracks, or swelling when subjected to -65°F. as specified in 4.4.4.2 and shall meet the applicable requirements of 3.8 while at this temperature. Cable and plugs shall meet the applicable requirements of 3.7 after completion of low temperature test while at 80°F ± 20°F.

3.9.3 Humidity. The cable assemblies shall withstand humidity of 95 ± 5 percent and maximum temperature of + 160°F. + 5°F. without evidence of damage or deterioration such as distortion, cracks, or swelling when tested as specified in 4.4.4.3. The cables and plugs shall meet the applicable requirements of 3.8 within one hour after removal from humidity chamber. Cables and plugs shall meet the applicable requirements of 3.7 after completion of humidity test while at + 80° + 20°F.

3.9.4 Transportation vibration. The cable assemblies shall withstand vibration for 15 ± 1 minutes in each of three mutually perpendicular axis without evidence of damage affecting function when tested as specified in 4.4.4.4. The frequencies and amplitudes shall conform to Table XXXVI. Following this test, the cables and plugs shall meet the applicable requirements of 3.6, 3.7 and 3.8.

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Table XXXVI

Frequency Range hertz (Hz)	Time Cycle (minutes)	Constant Acceleration or Double Amplitude
8-55-8	7.5 ± 0.5	0.060 ± 0.006 inch
55-500-55	7.5 ± 0.5	10 ± 1 g

3.9.5 Operational vibration. The cable assemblies shall withstand without evidence of damage or affecting functioning a programmed sequence of band-limited white Gaussian noise and sine wave vibration when tested as specified in 4.4.4.5. The total time of test shall be 210 seconds; 70 seconds in each of the three mutually perpendicular axis shown in Figure 1. The complex vibration test will consist of the following five sections programmed continuously:

a. White Gaussian noise at a level of 22 g root mean square (rms) band-limited between 15 and 1500 Hz for one second.
b. A 4 g rms swept sine wave superimposed on 7.5 g rms of white Gaussian noise band-limited between 15 and 1500 Hz for 15 seconds. The sine wave shall be swept from 800 to 1300 Hz at a rate increasing directly with frequency, one such sweep taking the full 15 seconds.
c. An 8 g rms swept sine wave superimposed on 15 g rms of white Gaussian noise band-limited between 15 and 1500 Hz for 15 seconds. The sine wave shall be swept from 1300 to 800 Hz at a rate decreasing directly with frequency, one such sweep taking the full 15 seconds.
d. A 4 g rms sine wave superimposed on 7.5 g rms of white Gaussian noise band-limited between 15 and 1500 Hz for 38 seconds. The sine wave shall be swept from 15 to 1500 Hz and back to 15 Hz at a rate increasing directly with frequency. One such cycle shall take the full 38 seconds.
e. White Gaussian noise at a level of 22 g rms band-limited between 15 and 1500 Hz for one second.

Following the vibration test, the cable assemblies shall meet the requirements of 3.6, 3.7 and 3.8.

3.9.6 Shock. The cable assemblies shall withstand shocks of 30 ± 5 gravity units (g) three times in the direction along the three mutually perpendicular axis shown in Figure 1, for a total of 9 shocks when tested as specified in 4.4.4.6.

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The duration of shock shall be 11 ± 2 milliseconds as measured between points which are 10 percent of the peaks. Following the tests, the assembly shall meet the requirements of 3.6, 3.7 and 3.8.

3.10 Test codes (see 3.5). The contractor shall provide data feed back on punch cards in conformance with Quality Assurance Instruction QAI-PA-1. The data required to be coded, and the code numbers, are specified in the applicable paragraph in this section.

3.11 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.12 Workmanship.

3.12.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.12.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.12.3 Burrs. Parts shall be free from burrs.

3.12.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.12.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.12.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

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4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see applicable drawing) for first article testing to the testing facility designated in the invitation for bids or request for proposal. Where a single contractor is manufacturing all or most of the assemblies of this specification, the contractor, upon approval by the Government, may submit three each of only the W5L, W10L and W44L cable assemblies as the total first article quantity.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table XXXVII. Tests need not be conducted in order specified in Table XXXVII.

Table XXXVII

First article inspection and tests

Inspection or Test	Dest	Non-Dest	Test Unit No. 1	2	3
1. Classification of characteristics (see 4.3.2.1)		X	X	X	X
2. Mating assembly (see 3.6)		X	X		X
3. Circuit resistance (see 3.4)		X	X	X	X
4. Insulation resistance (see 3.8)		X	X	X	X
5. Shock (see 3.9.6)	X		X		
6. Transportation vibration (see 3.9.4)		X	X		
7. Operational vibration (see 3.9.5)	X				X
8. Humidity (see 3.9.3)	X			X	X
9. High temperature (see 3.9.1)		X	X	X	
10. Low temperature (see 3.9.2)		X		X	X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

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4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table XXXVIII and Table XXXIX), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic are performed.

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Table XXXVIII
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table XXXIX
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Cable assemblies W1L, W2L, W3L, W5L, W6L, W10L, W44L, W25L, 7W2, T-4138-1, T-4138-2, T-4138-3, T-4138-4, and T-4138-5 before molding.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Any component missing, damaged or loose	Visual
	102. Presence of chips or foreign matter	Visual
	103. Improper soldering	Visual
	104. Improper connector	Visual
Minor:		
	201. General workmanship	Visual
	202. One half inch minimum overlap of molding compound of outer sheathing	Scale

4.3.2.1.2 Cable assemblies W1L, W2L, W3L, W5L, W6L, W10L, W44L, W25L, 7W2, T-4138-1, T-4138-2, T-4138-3, T-4138-3, T-4138-4 and T-4138-5 after molding.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Categories	Characterisitcs	Method of Inspection
Major:		
101.	Lead wire not completely covered by molded rubber	Visual
102.	General damage	Visual
103.	Incorrect length	Scale
104.	Connector in correct orientation	Visual
*105.	Improper hieght of "male" terminals (pins of the same diameter must all appear to be within 1/32 inch in height relative to the adjacent pins)	Visual/Gage
**106.	Improper depth of (female socket contact spring appear to be within 1/32 inch in depth relative to the adjacent sockets)	Visual/Gage
107.	Objective evidence relative to material improper or missing (see 3.1)	Visual
108.	Serialization improper or missing (see 3.4)	Visual

*In the event a pin does not appear to be within 1/32 inch in height relative to the adjacent pin, a gage (see 4.3.4) shall be used to determine the variation from the normal height. Reject any pin that varies by more than 0.032 inch maximum from the normal pin height of the individual connector. The normal height of the individual connector pin shall be determined by calculating the average height from measurements taken of pins adjacent to a pin that appears defective. The forward metal edge of the connector shall be utilized as the reference surface for all measurements.

**In the event a socket insert does not appear to be within 1/32 inch in depth relative to an adjacent socket, a gage (see 4.3.4) shall be used to determine the variation from the normal depth. Reject any insert that varies by more than 0.032 inch maximum from the normal socket insert depth of an individual connector. The normal depth of an individual connector socket shall be determined by calculating the average depth from measurements taken of sockets adjacent to the socket that appears defective. The forward metal edge of the connector shall be utilized as the reference surface for all measurements.

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Minor:

201.	Incorrect or illegible cable and connector markings	Visual
202.	Poor bond between molded rubber and tubing and/or connector shell	Visual
203.	Voids and air pockets greater than 1/8" diameter in rubber	Visual
204.	Workmanship (see 3.12)	Visual

4.3.2.1.3 Plug assemblies. - check plug - 2P6, fire plug - 8830734, and safe plug - 8837033, before potting

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Presence of chips or foreign matter	Visual
102.	Improper soldering	Visual
103.	Improper connector	Visual
Minor:		
201.	General workmanship	Visual

4.3.2.1.4 Plug assemblies. - check plug - 2P6, fire plug - 8830734, and safe plug - 8830733, after potting.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Objective evidence relative to material improper or missing (see 3.1)	Visual
102.	Serialization improper or missing (see 3.4)	Visual
Minor:		
201.	Incorrect or illegible marking	Visual
202.	Insufficient potting	Visual
203.	Bare spot in protective coating except for slight scratches	Visual
204.	Workmanship	Visual

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4.3.2.1.5 Cable assemblies. - Fuze: W7L, W8L and W9L before potting.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Any component missing, damaged or loose	Visual
102.	Presence of chips or foreign matter	Visual
103.	Improper soldering	Visual
104.	Improper connector	Visual
105.	Improper configuration	Visual
106.	Improper length	Scale
107.	General workmanship	Visual
Minor:	None defined.	

4.3.2.1.6 Cable assemblies. - Fuze: W7L, W8L and W9L after potting, but prior to assembling caps at potted joints.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Insufficient potting compound	Visual
102.	Objective evidence relative to material improper or missing (see 3.1)	Visual
103.	Serialization improper or missing (see 3.4)	Visual
Minor:		
201.	Workmanship (see 3.12)	Visual

4.3.2.1.7 Cable assemblies. - fuze: W7L, W8L and W9L after assembling of caps after potting.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Any component missing, loose or damaged	Visual

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4.3.2.1.8 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation, packaging and packing in accordance with MIL-STD-1169	Visual
102.	Marking in accordance with MIL-STD-1169	Visual
Minor:		
201.	Workmanship improper (see 3.12)	Visual

4.3.3 Testing. All tests shall be performed. Tests need not be performed in the order specified below:

Test	Test Classification
a. Mating assembly	Major
b. Circuit resistance	Major
c. Insulation resistance	Major
d. Low temperature	Major
e. Humidity	Major
f. Transportation vibration	Major

Of these tests, e, and f, are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Insulation resistance, and circuit resistance. Each cable assembly and plug shall be tested in accordance with 4.4.2 and 4.4.3. If any item fails to comply with 3.7 and 3.8, the item shall be rejected and removed from the lot.

4.3.3.2 Low temperature, humidity, and transportation vibration. A sample shall be selected at random in accordance with 4.4.4.2 through 4.4.4.4. If any item fails to comply with 3.9.2 through 3.9.4, the lot shall be rejected.

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Table XXXX

Lot Size	Sample Size
2-8	1
9-180	2
181-500	3
501-800	5
801-1300	7

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8854164). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Mating assembly. The cable assembly connectors will be mated to a fixture equivalent to the terminations they are to be used on.

4.4.2 Insulation resistance. The insulation resistance between each terminal shall be measured with a megohmmeter, or Ordance approved test circuit device, at 500± 50 volts direct circuit (dc). The insulation resistance shall not be less than 20 megohms as specified in 3.8. Any circuit exhibiting an insulation resistance of less than 20 megohms shall not be rejected until the reading has been verified by the application of the specified voltage for 60 ± 5 seconds. If, at the end of this time period, the insulation

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resistance has increased to at least 20 megohms, the circuit shall be accepted.

4.4.3 Circuit resistance. The resistance between the pins listed in 3.7 shall be measured with a Government approved electrical resistance tester. The resistance shall not exceed the value specified in 3.7.

4.4.3.1 Diode characteristics. Diodes shall be tested for forward voltage drop when the current specified in 3.8.1 is applied in the forward direction for not more than 15 seconds. Reverse current shall be determined by the voltage drop across a series resistor when the voltage specified in 3.8.1 is applied in the reverse direction for no more than 15 seconds.

4.4.4 Environmental testing (see 3.9).

4.4.4.1 High temperature. The cable assemblies with their connectors covered by potted mating connectors or environmental type mating connectors shall be placed in a temperature chamber and the temperature raised to + 160°F., for a period of 4 hours or until the temperature of the cable stabilizes (see 6.4). While at this temperature, the cable assemblies shall be tested as specified in 4.4.2 and shall meet the requirements specified in 3.8. Cables and plugs shall meet the requirements of 4.4.3 and 3.7 after completion of high temperature test and return to 80°F. ± 20°F.

4.4.4.2 Low temperature. The cable assemblies with their connectors covered by potted mating connectors or equivalent type mating connectors shall be placed in a test chamber and temperature lowered to -65°F. for a period of 4 hours or until the temperature of the cable assembly stabilizes (see 6.4). While at this temperature, the cable assemblies shall be tested as specified in 4.4.2 and shall meet the requirements of 3.8. Cables and plugs shall meet the requirements of 4.4.3 and 3.7 after completion of low temperature test and return to 80°F. ± 20°F.

4.4.4.3 Humidity. The cable assemblies with their connectors covered by potted mating connectors or environmental type mating connectors shall be placed within the test chamber and shall be subjected to the humidity test cycle as specified in MIL-E-5272, 4.4, Procedure I. All temperatures shall be maintained at +5°F. Three cycles shall be performed consecutively. The ph value shall be adjusted by the use of chemically pure hydrochloric acid or chemically pure sodium hydroxide. Upon completion of the last cycle, the cable assemblies shall be removed from the test chamber and, with the moisture removed by wiping, or blowing with compressed air or nitrogen, shall comply with the requirements specified in 3.9.3.

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4.4.4.4. Transportation vibration. The cable assemblies shall be mounted to a suitable shake table. This will be accomplished in the following manner: The cables shall be mounted to a fixture by attaching each cable connector to a mating connector mounted on the fixture, (see Figure 1). Spacing of the connectors on the fixture need not simulate final cable application position. Vibration shall be performed in each of three mutually perpendicular axis of the cable connectors. The accelerometer which indicates the vibration level shall be mounted as near the mounting fixture as is practical. Frequencies, amplitude and time of vibration shall conform to 3.9.4.

4.4.4.5 Operational vibration. The cable assemblies with no protective packaging shall be attached to a suitable shake table as specified in 4.4.4.4. The vibration levels shall be observed on the shake table as near the supporting bracket as is practical.

4.4.4.5.1 The standard noise tape. The standard noise or complex wave tape for this test shall have the characteristics as listed in the following table:

Time (sec)	Type of Signal	Ratio of Test Level To Calibration Level	RMS g Level
0-20	Noise (calibration)(1)	1.	Set gain to give 2 g rms
20-30	None	0	0
30-31	Noise (1)	11.	22.
31-46	Noise (1) plus sinusoid (2)	4.25	8.5
	Noise alone	3.75	7.5
	Sinusoid alone	2.	4.
46-61	Noise (1) plus sinusoid (3)	8.5	17.
	Noise alone	7.5	15.
	Sinusoid alone	4.	8.
61-99	Noise (1) plus sinusoid (4)	4.25	8.5
	Noise alone	3.75	7.5
	Sinusoid alone	2.	4.
99-100	Noise (1)	11.	22.

(1) Band-limited (15-1500 Hz) white Gaussian Noise.
(2) Sine wave swept from 800 to 1300 Hz at a rate increasing directly with frequency. One full sweep in 15 seconds.
(3) Sine wave swept from 1300 to 800 Hz at a rate decreasing directly with frequency. One full sweep in 15 seconds.
(4) Sine wave swept from 15 to 1500 Hz at a rate increasing directly with frequency. One full cycle in 38 seconds.

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For creating the random noise, the output of a General Radio random noise generator, or its equivalents, may be used if proper care is taken to insure the correct amplitude distribution of the signal. For testing the noise generator or any other associated equipment the use of an amplitude distribution analizer is suggested. For insuring correct noise bandwidth, a filter with an asymptotic slope of at least 24 Db per octave and 3 Db points at 15 and 1500 Hz shall be considered acceptable.

4.4.4.5.2 Method and procedure with the standard noise tape. Before applying the tape, the transfer function or frequency response of the entire shaking system shall be as flat as is consistent with the present state of the art. Plus or minus 3 Db or better over the band from 15 to 1500 Hz shall be considered as an objective in this respect. After proper equalization of the system, the standard noise tape is applied. The overall gain of the shake system is then adjusted so that with the noise calibration signal the g level listed in 4.4.4.5.1 is obtained. With the same gain setting, the tape is run for the remainder of the test and the g levels obtained will conform to the levels listed in the table of 4.4.4.5.1.

4.4.4.6 Shock. The cable assemblies shall be mounted on a suitable shock producing device. This will be accomplished in the following manner: The cables shall be mounted to a fixture by attaching each cable connector to a mating connector mounted on the fixture, (see Figure 1). The length of cable between the connectors shall not be attached to the fixture, the shock machine, or any stationary support, but will hang free. Spacing of the connectors on the fixture need not simulate final cable application position. The cable assemblies shall be chocked as specified in 3.9.6.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The cable assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The cable assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

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5.2.1 Levels A and C. The cable assembly shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 3.10 and 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-I.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Special first article conditions. First article samples will be required unless either of the following conditions exist:

a. An uninterrupted production of the identical item by the same manufacturer.
b. Ninety days have not elapsed since the identical item was produced by the same manufacturer presently under contract. Testing under Research and Development contracts shall not be cause for waiver or suspension of initial production testing.

6.6 Supersession data. This specification supersedes the requirements of Purchase description PA-PD-2075 (Rev.1), dated 31 August 1965.

Custodian:
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Preparing activity:
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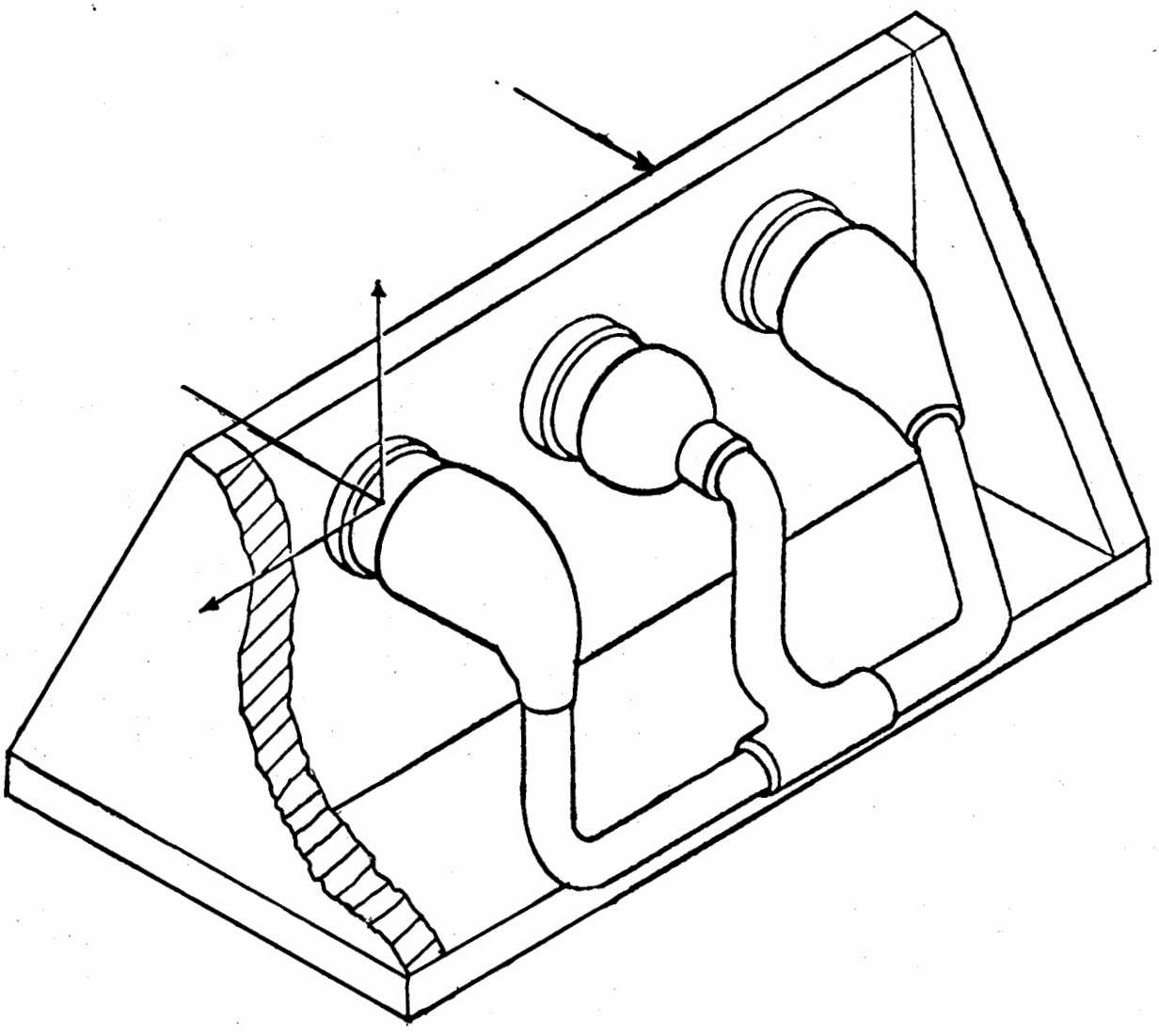


Figure 1 - Typical mounting of cable for vibration and shock

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